

YENEPOYA INSTITUTE OF ARTS, SCIENCE, COMMERCE AND MANAGEMENT

Final Project Report

on

Visual Threat Hunting for Manufacturing Networks

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1.1 Project Overview

The Manufacturing Threat Hunter is an AI-powered cybersecurity dashboard specifically designed for industrial manufacturing environments. Unlike traditional IT security tools, this system understands the unique characteristics of Operational Technology (OT) networks, including Programmable Logic Controllers (PLCs), Human-Machine Interfaces (HMIs), and Supervisory Control and Data Acquisition (SCADA) systems.

Core Problem Being Solved: Manufacturing networks face increasing cyber threats, but standard security tools often fail because they cannot distinguish between normal industrial traffic patterns (like batch processing data bursts) and actual attacks. This tool provides manufacturing-specific threat detection using four complementary machine learning models.

Development Context: The project is built entirely in Python using open-source libraries, making it accessible for small to medium manufacturing facilities that cannot afford enterprise security solutions. The dashboard-first approach prioritizes usability for plant operators who may not have deep cybersecurity expertise.

Repository Contents: The GitHub repository contains 2 commits (initial commit on April 27, 2026, and an update on April 29, 2026). Key files include app.py (main application), utils/ directory containing detector, preprocessing, visualizer, and report modules, plus high-level and low-level design documents in PDF format.

1.2 Key Capabilities

Multi-Model AI Detection: The system employs four distinct machine learning models:

- **Isolation Forest:** Rapidly isolates anomalies by randomly partitioning data. Works like finding a stranger in a crowd by asking yes/no questions.

- **Local Outlier Factor (LOF):** Compares each data point against its neighbors to find local density deviations. Like spotting someone behaving differently compared to their immediate surroundings.
- **K-Means Clustering:** Groups similar traffic patterns into clusters, then flags points that don't fit well into any cluster.
- **Random Forest:** Uses supervised learning to recognize previously seen attack patterns based on labeled examples.

Real-Time Monitoring: The live mode simulates network traffic streams, updating threat scores dynamically. This demonstrates how the system would behave when connected to actual network taps or industrial switches via port mirroring.

Predictive Forecasting: Using linear regression, the tool predicts traffic volume for the next hour. This enables proactive security measures, such as automatically increasing detection sensitivity during predicted high-risk windows.

Professional Reporting: Produces PDF security reports suitable for management review and audit compliance, plus CSV logs for detailed forensic analysis.

1.3 Critical Findings

Strengths Identified:

1. **Model Diversity:** Using four different detection approaches significantly reduces blind spots. Each model catches different attack patterns:
 - Isolation Forest captures global statistical outliers
 - LOF detects local anomalies that global methods miss
 - K-Means identifies new cluster formations indicating novel attack patterns

- Random Forest recognizes known attack signatures with high accuracy
2. **Manufacturing Relevance:** The detection logic (volume spikes, connection frequency, heuristics) directly addresses common manufacturing threats like:
 - DDoS attacks disrupting production schedules
 - Data exfiltration stealing intellectual property (product designs, recipes)
 - Brute force attempts on HMI and PLC access
 3. **Usability Focus:** The Streamlit interface provides immediate visual feedback without requiring command-line expertise. Plant operators can see threat scores color-coded (green/yellow/red) and understand risk levels instantly.

Gaps Identified:

1. **No Live Packet Capture:** The system only analyzes uploaded CSV files. To be production-ready, it needs integration with tools like libpcap, Scapy, or industrial protocol dissectors (Modbus, Profinet, EtherNet/IP).
2. **Simulated Live Mode:** The real-time monitoring feature generates synthetic data rather than processing actual network traffic. This is suitable for demonstration but not operational deployment.
3. **Missing Protocol Support:**
 - No OPC-UA (industrial standard for data exchange)
 - No MQTT (common in IIoT deployments)
 - No DNP3 (power grid and manufacturing protocol)
 - No Modbus TCP parsing
4. **Model Limitations:**

- Random Forest requires labeled attack data which may not exist for novel threats
- Models appear to retrain on each upload rather than incrementally learning, causing performance issues with large datasets

Recommendation Summary: The tool is excellent as a **proof-of-concept, educational resource, or post-incident forensic analyzer**. For production deployment, it requires integration with real-time packet capture and industrial protocol parsers. Organizations with mature security teams could use it as a supplemental analytics layer for CSV logs exported from existing security tools.

Chapter 2: Industrial Cybersecurity Landscape

2.1 Manufacturing Cyber Threats Evolution

Historical Context (1990s-2000s): Manufacturing networks were air-gapped (physically isolated from the internet). Security focused on physical access controls. Most threats were accidental (disgruntled employees, operational errors) or came via infected USB drives (Stuxnet being the most famous example in 2010).

Current Threat Landscape (2015-Present): Several trends have increased risk dramatically:

1. **Industry 4.0 and IIoT:** Manufacturing facilities now connect production networks to enterprise IT systems, cloud platforms, and partner networks for predictive maintenance, supply chain integration, and remote monitoring. Each connection creates a potential attack pathway.
2. **Ransomware Proliferation:** Manufacturing has become a top ransomware target because disrupting production causes immediate financial losses, forcing companies to pay. Examples include:

- 2021: Colonial Pipeline (not manufacturing but similar OT environment)
 - 2022: Toyota suspended production at 14 plants due to supply chain attack
 - 2023: Dole Food Company shut down North American production
3. **Nation-State Attacks:** Manufacturing is critical infrastructure in many countries. Threat actors steal intellectual property (chip designs, aerospace technology) or conduct sabotage:
- TRITON malware targeted safety instrumented systems in a petrochemical plant
 - Havex malware scanned for ICS devices and exfiltrated data

Unique Manufacturing Challenges:

- **Legacy Systems:** Many factories run equipment from 1990s or earlier that cannot be patched or secured with modern tools
- **Real-Time Requirements:** Security scanning can't introduce latency that disrupts production cycles (e.g., a robot stopping mid-operation)
- **Long Equipment Lifespan:** Manufacturing equipment operates for 15-25 years, far longer than typical security technology refresh cycles
- **Limited Security Expertise:** Plant IT teams may lack OT-specific security training

2.2 OT vs IT Security Differences

Aspect	IT Security	OT/Manufacturing Security
Primary Goal	Confidentiality (protect data)	Safety and availability (keep production running)

Aspect	IT Security	OT/Manufacturing Security
Critical Impact	Data breach, reputation damage	Physical damage, human injury, production loss
Patch Cadence	Monthly or as needed	Annual or never (requires validation downtime)
Network Architecture	Enterprise IT: segmented but connected	Often flat networks due to legacy protocols
Authentication	Strong (passwords, MFA)	Weak or non-existent (shared passwords, hardcoded)
Security Monitoring	Yes, standard practice	Rare or non-existent
Response Time	Minutes to hours	Seconds to milliseconds
Lifespan	3-5 years	15-25 years

Why OT Security is Harder:

First, manufacturing networks prioritize **deterministic behavior**. A security tool that adds random latency (e.g., scanning every packet) could cause a robot to miss a timing window, potentially damaging products or injuring workers. The Manufacturing Threat Hunter addresses this by operating offline on CSV logs rather than inline packet inspection.

Second, **protocols are different**. IT uses TCP/IP well-understood protocols (HTTP, SMTP, DNS). Manufacturing uses protocols like Modbus (no

authentication, easily spoofed), Profinet (requires specialized parsing), and proprietary protocols from vendors like Siemens, Rockwell Automation, and Mitsubishi.

Third, **risk tolerance is different**. In IT, a false positive (alerting on benign traffic) causes analyst annoyance. In OT, a false positive could trigger a production shutdown "just to be safe," costing \$10,000+ per minute in some facilities.

2.3 Attack Vectors in Manufacturing

External Attack Vectors:

1. **Remote Access Tools:** Many manufacturers allow vendors to remotely access systems for maintenance. These connections often use weak VPNs, unpatched remote desktop software, or vendor-specific backdoors. Example: An HVAC vendor's compromised laptop connected to the production network.
2. **Supply Chain Compromise:** Attackers target software updates, third-party components, or suppliers. Example: NotPetya ransomware spread via Ukrainian accounting software, eventually impacting Maersk shipping (manufacturing logistics) and Merck pharmaceutical production.
3. **Phishing:** Employees receive emails that install malware. Once inside the IT network, attackers pivot to OT systems, often using default credentials on HMIs and PLCs.

Internal Attack Vectors:

1. **Disgruntled Employees:** Former or current employees with legitimate access. Manufacturing has seen cases of workers deleting recipes, altering temperatures to damage products, or disabling safety systems.
2. **Infected USB Drives:** Maintenance technicians plug in laptops or USB drives that have been infected elsewhere. The Stuxnet worm famously spread via USB drives at an Iranian uranium enrichment facility.

3. Bring Your Own Device (BYOD): Plant

managers using personal tablets to monitor production dashboards, potentially carrying malware from home networks.

Attack Targets in Manufacturing:

- **Disrupt Production:** Ransomware or denial-of-service attacks stop assembly lines, causing revenue loss
- **Steal Intellectual Property:** Product designs, chemical formulas, CNC programming files
- **Cause Physical Damage:** Alter PLC parameters to destroy equipment (e.g., overspeed motors, disable cooling)
- **Safety System Compromise:** Disable emergency shutdown or safety interlocks (extremely dangerous)

2.4 Regulatory Requirements

NIST SP 800-82 (Guide to Industrial Control Systems Security):

This is the foundational US standard. Key requirements relevant to Manufacturing Threat Hunter:

- **Continuous monitoring** of boundary and internal network traffic (Chapter 6.2.6). The tool's CSV upload and real-time simulation partially address this for post-hoc analysis.
- **Anomaly detection** using statistical, signature-based, and behavioral methods (Section 6.1.3). The four-model ensemble directly implements this requirement.
- **Incident response** with forensic logging (Section 5.3.6). The PDF and CSV exports support this.

IEC 62443 Series (International Standard):

This is becoming mandatory in many jurisdictions. Key parts:

- **IEC 62443-2-1 (Security**

Management): Requires security monitoring and continuous improvement. The reporting and analytics features address this.

- **IEC 62443-3-3 (System Security Requirements):** Requires monitoring for "unusual events and anomalies" (SR 6.2). The threat scoring engine directly meets this.

Industry-Specific Regulations:

- **CFATS (Chemical Facilities):** US chemical manufacturers must monitor for cyber threats that could cause release of hazardous materials.
- **NERC CIP (Electric Power):** Applies to power generation (including some manufacturing). Requires logging and audit capabilities.
- **GDPR (Europe):** Manufacturing often processes personal data (employee records, customer orders). Security monitoring must protect this data.

Compliance Gaps in Current Tool:

- No authentication or access logging (required by all frameworks)
- No encryption of stored data (CSV files are plaintext)
- No audit trail of who generated reports or modified configurations

2.5 Market Gap Analysis

Current Commercial Solutions:

Product	Price Range	Strengths	Weaknesses
Nozomi Networks Guardian	\$50k+/year	Real-time OT protocol parsing, AI detection	Expensive, complex deployment

Product	Price Range	Strengths	Weaknesses
Dragos Platform	\$100k+/year	Threat intelligence, analyst support	Enterprise-focused
Claroty	\$75k+/year	Deep protocol dissectors	Requires dedicated team
Microsoft Defender for IoT	\$20k+/year	Cloud integration, easy setup	Requires Azure, limited OT protocols

Gap Filled by Manufacturing Threat Hunter:

The open-source, tool targets **small to medium manufacturers** (less than \$500M revenue) who cannot afford commercial solutions. These facilities:

- Have IT budgets under 500k *total* (*cannot spend* 50k+ on security)
- May not have dedicated security staff
- Need basic visibility without complex deployment
- Often already collect network logs for compliance

Comparison to Similar Open-Source Tools:

Tool	Focus	OT Support	ML Detection	Dashboard
Zeek (formerly Bro)	Network monitoring	Limited (requires scripts)	No, only signatures	Basic

Tool	Focus	OT Support	ML Detection	Dashboard
Suricata	IDS/IPS	Some ICS rules	No	Command-line
Wazuh	SIEM	Limited	Basic (single model)	Yes
Manufacturing Threat Hunter	OT threat detection	Generic (CSV-based)	Yes (4 models)	Excellent

Opportunity for Improvements:

1. **Add OT Protocol Parsers:** Modbus, Profinet, EtherNet/IP would make the tool directly usable
2. **Develop Pre-trained Models:** The repository doesn't include pre-trained model files (models/*.pkl appear empty). Provide models trained on public ICS datasets (IEC 60870-5-104, Modbus)
3. **Create Docker Deployment:** Simplify installation for non-Python experts

Chapter 3: System Requirements

3.1 Hardware Requirements

Minimum Configuration (Development/Lab):

- **CPU:** 2 cores, 2.0 GHz (Intel Core i3 or AMD equivalent)

- **RAM:** 4 GB minimum, 8 GB recommended

- **Storage:** 2 GB for application + variable for CSV files
- **Network:** No real-time capture required (file-based)

Why These Minimums: The four ML models require memory to load and process data. With 4 GB RAM, maximum CSV size is approximately 500 MB (based on typical pandas DataFrame memory usage of 5-10x file size). The Random Forest model alone may use 500 MB-1 GB depending on tree depth.

Recommended Configuration (Production Testing):

- **CPU:** 4 cores, 2.5+ GHz (Intel Core i5/i7 or AMD Ryzen 5/7)
- **RAM:** 16 GB
- **Storage:** 20 GB SSD (for faster CSV reads/writes)
- **GPU:** Not required (scikit-learn doesn't support GPU significantly)

Scaling Estimates:

- With 16 GB RAM, can process CSVs up to 2-3 GB
- Processing time approximately:
 - 100 MB CSV: 10-30 seconds
 - 1 GB CSV: 2-5 minutes

3.2 Software Dependencies

Operating Systems (Tested):

- Windows 10/11 (requires Python 3.8+)
- Ubuntu 20.04/22.04 (recommended for performance)
- macOS 11+ (Big Sur or newer)

Python Version:

- Requires Python 3.8, 3.9, 3.10, or 3.11
- Not tested on Python 3.12+ (some package compatibility issues possible)

Core Dependencies (from requirements.txt):

text

```
streamlit==1.28.0      # Dashboard framework
pandas==2.0.3          # Data manipulation
numpy==1.24.3          # Numerical operations
scikit-learn==1.3.0    # All ML models
matplotlib==3.7.2      # Static charts
seaborn==0.12.2        # Statistical visualizations
plotly==5.15.0         # Interactive charts
fpdf==2.7.4            # PDF generation
joblib==1.3.2          # Model serialization
```

Installation Time: Approximately 5-10 minutes including download of dependencies (depends on internet speed). Total install size including dependencies: ~500 MB.

Virtual Environment Required: Yes, strongly recommended to avoid conflicts. The README provides commands for venv creation.

3.3 Network Prerequisites**For CSV Upload Mode (Only Mode in Current Version):**

- No network connectivity required to run the tool
- CSV files can be transferred via USB, network share, or any method
- Dashboard accessible at <http://localhost:8501> (default Streamlit port)

- Can expose to network by changing Streamlit settings (--server.address 0.0.0.0)

For Production Deployment (If Enhanced):

- Port mirroring/SPAN port configuration on industrial switches
- Access to network taps near core switch or production cell
- Bandwidth requirements: 1-10 Mbps typical for OT traffic

Firewall Rules (If Web Access Needed):

- Allow TCP 8501 inbound for dashboard access
- Restrict to management network (not exposed to production floor)
- No outbound internet required (unless for updates)

3.4 Data Format Specifications

CSV Column Requirements (Based on Code Analysis):

The temp_upload.csv file in the repository provides the schema. While not explicitly documented, typical network log CSV includes:

Column Name	Data Type	Description	Example
timestamp	datetime	When traffic occurred	2026-04-27 14:23:05
src_ip	string	Source IP address	192.168.1.100
dst_ip	string	Destination IP address	10.0.0.50
protocol	string	TCP/UDP/ICMP	TCP

Column Name	Data Type	Description	Example
src_port	integer	Source port number	54321
dst_port	integer	Destination port number	502 (Modbus)
packet_size	integer	Bytes in packet	1460
packet_count	integer	Number of packets in flow	150

Required Minimum Fields for Detection to Work:

- packet_count or bytes_transferred (volume factor)
- timestamp (temporal features)
- src_ip and dst_ip (connection frequency)

Data Quality Requirements:

- No missing values in required columns (will cause preprocessing errors)
- Timestamps should be in consistent format (YYYY-MM-DD HH:MM:SS)
- IP addresses in standard IPv4 dotted decimal format

Sample Data Generation:

If you don't have real data, you can:

1. Use tcpdump or Wireshark to capture network traffic
2. Convert .pcap to CSV using tools like Zeek (formerly Bro)
3. Generate synthetic data using included simulation in live mode

File Size Limits:

- No hard limit in code, but practical limit based on memory
- For 8 GB RAM: keep CSV under 500 MB
- For large files, consider splitting by time window